

Changes in running economy and attainable maximal oxygen consumption in response to prolonged running: The impact of training status

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Abstract

During prolonged running at moderate-to-high intensity, running economy (RE) deteriorates and attainable maximal oxygen consumption (VO_{2max}) decreases. Whether these changes appear similarly in trained and untrained runners exercising at the same relative intensity is not clear. We recruited 10 trained runners (TR) and 10 active adults (AA), and compared RE and attainable VO_{2max} before and after 1 h of running at 70% of VO_{2max} . Submaximal VO_2 increased more ($p=0.019$) in AA ($0.20 \pm 0.13 \text{ L min}^{-1}$) than in TR ($0.07 \pm 0.05 \text{ L min}^{-1}$). Attainable VO_{2max} decreased in AA ($-0.21 \pm 0.15 \text{ L min}^{-1}$, $p=0.002$), but remained unchanged in TR ($-0.05 \pm 0.10 \text{ L min}^{-1}$, $p=0.18$). Relative intensity (i.e., $VO_2/\text{attainable } VO_{2max}$), increased more ($p=0.001$) in AA ($8.3 \pm 4.4\%$) than in TR ($2.6 \pm 1.9\%$). These results demonstrate that the ability to resist changes in RE and VO_{2max} following prolonged running is superior in trained versus untrained runners, when exercising at the same relative intensity.

KEYWORDS

durability, endurance performance, physiological resilience, VO_{2max} , work economy

1 | INTRODUCTION

Maximal oxygen consumption (VO_{2max}) and running economy (RE) are well-known predictors of running performance.¹ In combination, these two factors account for as much as 85%–94% of variability in running performance.^{2,3} Most commonly, VO_{2max} and RE are measured in a rested condition, after a short warm up procedure designed to optimize the result. During long-lasting running competitions, however, measurements obtained in a rested condition may not be representative throughout the competition, as the athlete starts to become fatigued.⁴

During prolonged running at a fixed velocity, the oxygen cost of running is shown to increase. In a group of highly trained male long distance runners, Xu and Montgomery⁵ observed 1.8 and $3.2 \text{ mL kg}^{-1} \text{ min}^{-1}$ increase in VO_2 , following 90 min of running at 65% and 80% of VO_{2max} , respectively. Similarly, in a group of male physical education students, Sproule⁶ observed 2.6 and $3.7 \text{ mL kg}^{-1} \text{ min}^{-1}$ increase in VO_2 , running 1 h at 70% and 80% of VO_{2max} , respectively.

The ability to reach VO_{2max} during running is also compromised following prolonged heavy running exercise. In a group of trained male runners, attainable VO_{2max}

	Trained runners (<i>n</i> = 10)	Active adults (<i>n</i> = 10)
Men/Women	6/4	6/4
Age, year	32 ± 8	30 ± 2
Body mass, kg	69 ± 8	79 ± 12 [#]
Height, cm	177 ± 7	176 ± 9
VO _{2max} (mL kg ⁻¹ min ⁻¹)	64.4 ± 6.9	49.8 ± 4.9 ^{##}
Personal best at 10km running (min)	35.2 ± 3.0	NA
Men/Women	33.1 ± 1.3/38.3 ± 1.6	NA
Years of running, year	9 ± 4	NA
Running distance, km week ⁻¹	69 ± 15	9 ± 5 ^{##}
Other endurance training, h week ⁻¹	3.4 ± 2.3	0.8 ± 1.0 ^{##}
High-intensity endurance training, sessions week ⁻¹	2.3 ± 0.4	0.9 ± 0.5 ^{##}

Note: Data are mean ± SD. [#]*p* < 0.05, ^{##}*p* < 0.01 significant difference between groups.

TABLE 1 Subject characteristics of trained runners and active young adults.

was reduced from 60.0 to 56.3 mL kg⁻¹ min⁻¹, when measured 30 min after completion of a half-marathon run.⁷ In agreement with this, De Vito et al.⁸ observed a 5 mL kg⁻¹ min⁻¹ reduction in running VO_{2max} (from 69 to 64 mL kg⁻¹ min⁻¹), immediately after the first two segments of a triathlon (1.5 km swimming + 32 km cycling). Exercising in the heat may further amplify decrements in attainable VO_{2max}.⁹ Importantly, the combination of reduced attainable VO_{2max} and increased VO₂ at a given velocity, may significantly increase the relative intensity of prolonged steady pace running.⁴

Recently, the term physiological resilience (or durability) has been introduced in the literature, to describe an athlete's ability to resist changes in physiological profiling attributes during prolonged exercise.^{4,10,11} In a series of studies, Clark et al.^{12–14} showed that cycling critical power is markedly reduced after 2 h cycling at 60%–65% of VO_{2max}. The reduction in critical power was accompanied by, but not correlated with, an increased O₂ cost of cycling. Contrary to the aforementioned studies of Dressendorfer⁷ and De Vito et al.,⁸ Clark et al.^{12–14} observed no change in cycling VO_{2max} after 2 h of biking, when assessed during 3 min of all-out effort. This discrepancy may rely on inherent differences between running and cycling exercise, however training status of the subjects may also be influential.

Despite being a relatively new area of research, physiological resilience has already been suggested to represent a fourth parameter of endurance exercise physiology,⁴ complementing the established model consisting of VO_{2max}, lactate threshold, and RE.¹⁵ To date, the literature seems to agree that relatively large differences in physiological resilience exist between individuals,^{4,10,12–14,16} however, evidence is scarce of which factors that actually determine this resilience. Perhaps most important, it is not

clear whether durability is different between trained and untrained subjects exercising at the same relative intensity; that is, if durability is a physiological trait that positively adapts to training, such as VO_{2max} and RE. In Clark et al.,^{12–14} critical power resilience did not correlate with any baseline parameter of cardiorespiratory fitness. On the other hand, field data from elite cycling suggests that the ability to produce high wattage in a fatigued state is a characteristic that separates professional road cyclists from elite under 23 and junior cyclists.¹⁷ The purpose of the present study was therefore to investigate whether the ability to resist changes in VO_{2max} and RE is different between trained and untrained runners, when tested before and after 1 h of running at 70% of VO_{2max}. Specifically, we hypothesized that (1) the ability to resist changes in RE would be superior in trained versus untrained runners, and (2) the ability to resist changes in VO_{2max} would be superior in trained versus untrained runners. As an inevitable consequence of Hypotheses 1 and 2, we also hypothesized that (3) the relative intensity, that is, VO₂/attainable VO_{2max}, would increase more in the untrained than in the trained runners.

2 | METHODS

2.1 | Subjects

A total of 20 healthy volunteers; 12 males and 8 females, aged 31 ± 6 years, participated in the study (Table 1). Ten subjects (6 males and 4 females) were trained runners (TR), and 10 subjects (6 males and 4 females) were active young adults (AA). AA reported to participate regularly in recreational sport and strength training, but did not perform endurance training regularly more than two times a week. AA

were recruited among University students and staff, as well as local recreational sport clubs. TR were recruited through result lists from local running competitions. Male TR had to be able to run 10 km in less than 36 min, and female TR had to be able to run 10 km in less than 40 min. All subjects reported to be familiar with treadmill running. Written informed consent was obtained from all subjects prior to participation, after having received detailed information about the study. The study was approved by the Norwegian Centre for research data (SIKT reg. 570843), and carried out in agreement with the latest revision of the Declaration of Helsinki. The study was conducted in accordance with Nord University's research ethical regulations. The regional ethical committee of health research reviewed the experimental protocol and confirmed that ethical approval for patient research was not required.

2.2 | Experimental design and physiological testing

Each subject attained 2 days of testing in the physiological laboratories at Nord University. All running was conducted on a calibrated treadmill (Woodway 4Front, Weil am Rhein, Germany). Running was conducted at 1% incline.¹⁸ Temperature in the laboratory was held at 20–22°C. Subjects were not allowed to use running shoes with carbon plates.

2.2.1 | Test day 1

On the first day of testing, subjects started with 5 min self-determined easy running, after which warm up blood lactate concentration ($[La^-]_b$) was established, using a portable lactate analyzer (Lactate Scout 4, EKF Diagnostics, UK). Thereafter, subjects conducted a RE and lactate profiling procedure, consisting of three to four work periods of 5-min running. For the first work period, velocity was set to a predicted intensity of ~50% of VO_{2max} ,³ based on the subjects self-reported or self-predicted race time at 10 km running. For each step the velocity was increased by 1–2 km h⁻¹, until $[La^-]_b \geq 4$ mmol L⁻¹, or the rating of perceived exhaustion ≥ 7 (RPE 1–10). For the last 3 min of each 5-min work period, VO_2 was measured using a Vyntus CPX with mixing chamber mode (Vyair medical, USA), to determine RE. Between each work period 1-min rest were given, in which a blood sample was drawn to determine $[La^-]_b$. After the RE profiling procedure, subjects were given 10 min active rest (walking or easy running on the treadmill). VO_{2max} was then determined during an incremental stepwise protocol at either 5% or 10% incline, based on the subjects' preference. By educated guessing, the initial velocity was set somewhat below the subjects'

predicted lactate threshold at the chosen incline, and velocity was thereafter increased by 1 km h⁻¹ every minute, until exhaustion, typically within 5–9 min. $[La^-]_b$ was determined 1 min after termination of the test. Heart rate (HR) was measured using a Polar Pacer heart rate monitor (Kempele, Finland).

2.2.2 | Test day 2

Using RE and VO_{2max} data from test day 1, the velocity corresponding to 70% of VO_{2max} was estimated. A linear regression was performed using VO_2 and running velocities from the steps in the lactate profiling procedure with $[La^-]_b < 3$ mmol L⁻¹. The point at which this regression line corresponded to 70% of VO_{2max} was selected as velocity for the 1-h running session. On the second day of testing, subjects started with 8 min progressive easy running, culminating in 2 min running at the predicted 70% of VO_{2max} velocity; 10 min warm up in total. VO_2 at the predicted 70% of VO_{2max} velocity was then established during a 5-min reference run at this velocity, followed by 1 min rest to determine $[La^-]_b$. Thereafter, the subject ran 1 h at the same velocity (hereafter referred to as *1 h running session*), during which VO_2 was measured every 15 min (i.e., between 12–15, 27–30, 42–45, and 57–60 min). Data from the 5-min reference run is referred to as “min 0” of the 1 h running session. For each of the 3-min assessment windows, the breathing mouthpiece used to measure VO_2 was inserted while running. Outside these assessment windows subjects did not use the mouthpiece, and were allowed to drink water. After 1 h the subject received a 2-min rest, in which $[La^-]_b$ and body mass were measured. Immediately after this, treadmill incline was set to 5% or 10% (the same incline used at test day 1), and the subject continued into the exact same VO_{2max} protocol as on test day 1, to assess attainable VO_{2max} after 1 h of running. One minute after termination of the VO_{2max} protocol, $[La^-]_b$ was obtained.

2.3 | Data analysis

2.3.1 | Maximal oxygen consumption

VO_2 measurements were sampled every 10 s. To determine VO_{2max} , the average of the six highest consecutive 10-second periods were used. Gas exchange threshold was determined by visually identifying the first disproportionate increase in VCO_2 versus VO_2 . This procedure was assisted by a computerized regression analysis tool for identification of exercise thresholds (SentrySuite software, Vyair Medical, USA), as described by Beaver, Wasserman and Whipp.¹⁹ At least 30 s at the final velocity

of the stepwise protocol had to be completed for this velocity to be considered *velocity at the end of VO_{2max} stepwise protocol*. If 30 s was not completed the former velocity was reported. As incline during the stepwise protocol was not identical between the groups (AA: 9 subjects used 5%, 1 subject used 10% incline; TR: 6 subjects used 5%, 4 subjects used 10% incline) this measure can only be used to quantify within subject changes in performance, between rested and fatigued condition.

2.3.2 | Running economy

At the submaximal speeds, VO_2 was sampled for a period of 3 min, of which the average of the last 90 s were used for calculation. RE refers to O_2 cost per kg body mass per meter ($mL kg^{-1} m^{-1}$), and was calculated as the average value from the submaximal speeds with $[La^-]_b < 3 mmol L^{-1}$. VO_2 refers to the submaximal VO_2 measured during the 1 h running session. Energy expenditure ($Kcal min^{-1}$) was estimated by multiplying VO_2 with the caloric equivalent to the corresponding RER value.

2.3.3 | Blood lactate

As blood lactate profiling was not a main research question in this study, a cut of at RPE ≥ 7 was introduced to ensure that the subsequent VO_{2max} testing was not compromised by exhaustion from the lactate profiling procedure. As a consequence of this the common lactate threshold value of $[La^-]_b = 4 mmol L^{-1}$ was not reached in all subjects. Results from lactate profiling are reported as velocity and % of VO_{2max} at warm up $[La^-]_b + 1.5 mmol L^{-1}$,^{20,21} and at 2 and 3 $mmol L^{-1}$ $[La^-]_b$.

2.3.4 | Body mass and fluid intake

Body mass was measured at three occasions during the study. At the start of test day 1, at the start of test day 2, and immediately after termination of 1 h running at 70% of VO_{2max} . The difference between the two measurements on test day 2, in addition to the magnitude of fluid intake, were used to calculate weight loss during the 1 h running session.

2.4 | Statistical analysis

Statistical analyses were performed using the software package SPSS version 29 (Chicago, IL, USA). Figures were made using Graph pad prism 10 (San Diego, CA, USA).

Student's independent t -tests were used to examine differences between groups in the rested condition. Paired t -tests were used to examine within group differences (from rested to fatigued condition). Independent t -tests were used to assess differences in changes between the groups (change from rested to fatigued condition). The Pearson test for linear regression was used to assess correlations. Significance level was set to $p > 0.05$. Data reported in tables and text are means $\pm$ SD.

3 | RESULTS

3.1 | Baseline differences between trained runners and active adults

Most of the measured physiological and performance variables reflected superior capacity and performance level in TR compared to AA (Table 2). Relative VO_{2max} was ~29% higher in TR than in AA ($p < 0.001$), and RE was 8% better ($p < 0.001$). Velocity at warm up $[La^-]_b + 1.5 mmol L^{-1}$ was ~41% higher in TR than in AA ($p < 0.001$). % of VO_{2max} at warm up $[La^-]_b + 1.5 mmol L^{-1}$ was not significantly different between the groups ($p = 0.22$). The % of VO_{2max} at gas exchange threshold was not different between the groups ($p = 0.32$).

3.2 | Changes in cost of running

During 1 h running, VO_2 ($L min^{-1}$) increased more in AA than in TR (AA: $6.9 \pm 3.9\%$ vs. TR: $2.3 \pm 1.5\%$, $p < 0.01$) (Figure 1, Table 3). Also HR (AA: $17.8 \pm 8.3\%$ vs. TR: $6.8 \pm 3.5\%$, $p < 0.001$), VE (AA: $26.8 \pm 17.5\%$ vs. TR: $5.5 \pm 6.0\%$, $p < 0.01$), and BF (AA: $51.8 \pm 26.9\%$ vs. TR: $14.1 \pm 13.7\%$, $p < 0.001$) increased more in AA than in TR (Figure 2).

3.3 | Changes in VO_{2max} , relative intensity and performance

VO_{2max} ($L min^{-1}$) decreased after 1 h running in AA ($-5.0 \pm 3.4\%$, $p < 0.01$), but remained unchanged in TR ($-1.2 \pm 2.4\%$, $p = 0.17$) (Table 4). The change in VO_{2max} was different between AA and TR ($p = 0.011$). The running velocity at the final stage of the VO_{2max} stepwise protocol decreased in AA ($-4.6 \pm 5.8\%$, $p < 0.05$), but remained unchanged in TR ($-1.3 \pm 3.2\%$, $p = 0.18$). In combination, the increase in VO_2 and the reduction in attainable VO_{2max} , resulted in a larger increase in relative intensity in AA than in TR (AA: $8.3 \pm 4.4\%$ vs. TR: $2.6 \pm 1.9\%$, $p = 0.001$) (Figure 3).

TABLE 2 Data from running economy and lactate profiling at test day 1.

	Trained runners (<i>n</i> = 10)	Active adults (<i>n</i> = 10)
RE (mL kg ⁻¹ m ⁻¹)	0.199 ± 0.010	0.216 ± 0.010 ^{##}
Warm up [La ⁻] _b (mmol L ⁻¹)	0.8 ± 0.3	1.3 ± 0.5 ^{##}
Velocity (km h ⁻¹) at [La ⁻] _b = 2 mmol L ⁻¹	14.8 ± 1.3	9.9 ± 1.2 ^{##}
VO ₂ (mL kg ⁻¹ min ⁻¹) at [La ⁻] _b = 2 mmol L ⁻¹	49.2 ± 4.3	35.3 ± 3.4 ^{##}
% of VO _{2max} at [La ⁻] _b = 2 mmol L ⁻¹	76.5 ± 3.4	71.3 ± 7.7
Velocity (km h ⁻¹) at warm up [La ⁻] _b + 1.5 mmol L ⁻¹	15.1 ± 1.5	10.7 ± 1.5 ^{##}
VO ₂ (mL kg ⁻¹ min ⁻¹) at warm up [La ⁻] _b + 1.5 mmol L ⁻¹	50.4 ± 5.4	38.0 ± 3.7 ^{##}
% of VO _{2max} at warm up [La ⁻] _b + 1.5 mmol L ⁻¹	78.3 ± 3.3	76.5 ± 6.5
Velocity (km h ⁻¹) at [La ⁻] _b = 3 mmol L ⁻¹	16.0 ± 1.5	10.9 ± 1.4 ^{##}
VO ₂ (mL kg ⁻¹ min ⁻¹) at [La ⁻] _b = 3 mmol L ⁻¹	53.0 ± 5.0	39.1 ± 3.8 ^{##}
% of VO _{2max} at [La ⁻] _b = 3 mmol L ⁻¹	82.4 ± 3.0	78.8 ± 7.1
Gas exchange threshold (mL kg ⁻¹ min ⁻¹)	39.4 ± 4.3	30.8 ± 3.8 ^{##}
Gas exchange threshold (% of VO _{2max})	61.2 ± 3.2	62.0 ± 3.9
1 h velocity (km h ⁻¹) (predicted 70% of VO _{2max})	13.7 ± 1.4	9.7 ± 1.3 ^{##}
% of Gas exchange threshold at 1 h velocity	111 ± 8	113 ± 6

Note: Data are mean ± SD. **p* < 0.05, ^{##}*p* < 0.01 significant difference between groups.

Abbreviations: [La⁻]_b, lactate concentration in blood; RE, running economy; VO_{2max}, maximal oxygen consumption.

3.4 | Weight loss and fluid intake

Weight loss (AA: 0.66 ± 0.30 kg vs. TR: 0.73 ± 0.28 kg, *p* = 0.60), and fluid intake (AA: 0.28 ± 0.26 L vs. TR: 0.37 ± 0.25 L, *p* = 0.46) during 1 h running were not different between AA and TR.

3.5 | Correlations

Across groups, ΔVO₂ (mL kg⁻¹ min⁻¹) correlated with Δvelocity at the end of VO_{2max} stepwise protocol (*R* = -0.45, *p* < 0.05), ΔHR (*R* = 0.61, *p* < 0.01), ΔVE

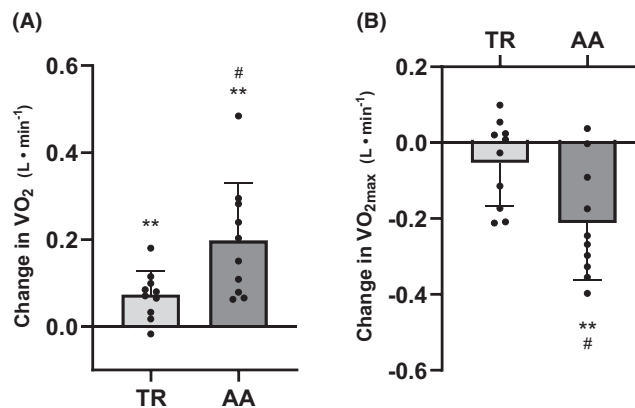


FIGURE 1 (A) Change in oxygen cost of running at predicted 70% of VO_{2max} velocity, from the beginning to the end of 1 h running. (B) Change in attainable VO_{2max} from the rested condition to 2 min after the end of 1 h running. Data are mean ± SD. *n* = 10 in both groups. **p* < 0.05, ***p* < 0.01 change within group from rested condition to the end of 1 h running. #*p* < 0.05 magnitude of change is different between groups.

(*R* = 0.64, *p* > 0.01), ΔBF (*R* = 0.71, *p* < 0.01). ΔVO_{2max} (mL kg⁻¹ min⁻¹) correlated with Δvelocity at the end of VO_{2max} ramp protocol (*R* = 0.54, *p* < 0.05), ΔHR (*R* = -0.63, *p* < 0.01), ΔVE (*R* = -0.55, *p* = 0.01), and ΔBF (*R* = -0.62, *p* < 0.01). Relative intensity at the end of 1 h correlated with Δvelocity at the end of the VO_{2max} ramp protocol (-0.73, *p* < 0.001), ΔHR (*R* = 0.67, *p* < 0.01), ΔVE (*R* = 0.67, *p* > 0.01), and ΔBF (*R* = 0.70, *p* < 0.01).

Within AA, relative intensity at the end of 1 h correlated with Δvelocity at the end of the VO_{2max} stepwise protocol (*R* = -0.715, *p* < 0.05). ΔVO₂ (mL kg⁻¹ min⁻¹) correlated with ΔBF (*R* = 0.62, *p* < 0.05). ΔVO_{2max} (L min⁻¹) correlated with ΔBF (*R* = -0.66, *p* < 0.05), and ΔHR (*R* = 0.63, *p* < 0.05).

Within TR, relative intensity at the end of 1 h correlated with Δvelocity at the end of the VO_{2max} stepwise protocol (-0.735, *p* < 0.05).

4 | DISCUSSION

This study sought to investigate whether physiological resilience during prolonged running is different between TR and AA. When tested before and after 1 h running at predicted 70% of VO_{2max} velocity, the main findings were that (1) VO₂ increased more in AA than in TR, (2) attainable VO_{2max} decreased in AA but not in TR, and (3) the relative intensity (i.e., VO₂/attainable VO_{2max}) increased more in AA than in TR. These results suggest that the ability to resist changes in RE and attainable VO_{2max} during prolonged running is superior in TR compared to AA. Arguably, physiological responses to prolonged exercise

TABLE 3 Data from 1 h running at the velocity corresponding to 70% of VO_{2max}

	0 min	15 min	30 min	45 min	60 min	Δ 0–15 min	Δ 0–30 min	Δ 0–45 min	Δ 0–60 min
Trained runners									
VO_2 ($L \cdot min^{-1}$)	3.12 ± 0.62	3.13 ± 0.62	3.17 ± 0.65	3.20 ± 0.63	3.19 ± 0.63	0.01 ± 0.03	0.05 ± 0.05	0.08 ± 0.07	0.07 ± 0.06
VO_2 ($mL \cdot kg^{-1} \cdot min^{-1}$)	44.8 ± 4.9	45.0 ± 5.1	45.5 ± 5.5**	46.0 ± 5.5**	45.9 ± 5.3**	0.2 ± 0.4	0.7 ± 0.7	1.2 ± 1.0	1.1 ± 0.8
Kcal min^{-1}	15.5 ± 3.0	15.5 ± 3.1	15.6 ± 3.2	15.7 ± 3.1*	15.7 ± 3.1*	0.0 ± 0.2	0.1 ± 0.2	0.3 ± 0.3	0.2 ± 0.3
RER	0.93 ± 0.03	0.92 ± 0.03	0.90 ± 0.02**	0.90 ± 0.02**	0.88 ± 0.02**	−0.01 ± 0.02	−0.03 ± 0.01	−0.03 ± 0.02	−0.05 ± 0.03
$[La^-]_b$	1.4 ± 0.5	—	—	—	1.4 ± 0.8	—	—	—	0.0 ± 0.8
Active adults									
VO_2 ($L \cdot min^{-1}$)	2.67 ± 0.48	2.77 ± 0.54**	2.81 ± 0.56**	2.83 ± 0.59**	2.87 ± 0.59**	0.10 ± 0.08**	0.14 ± 0.11 [#]	0.16 ± 0.12	0.20 ± 0.13 [#]
VO_2 ($mL \cdot kg^{-1} \cdot min^{-1}$)	34.0 ± 3.1	35.2 ± 3.5**	35.8 ± 3.5**	36.0 ± 4.0**	36.4 ± 4.0**	1.2 ± 0.9 [#]	1.7 ± 1.3 [#]	2.0 ± 1.4	2.4 ± 1.4 [#]
Kcal min^{-1}	13.2 ± 2.4	13.7 ± 2.7**	13.9 ± 2.8**	14.0 ± 3.0**	14.1 ± 3.0**	0.5 ± 0.4 [#]	0.7 ± 0.5 [#]	0.8 ± 0.6 [#]	0.9 ± 0.6 [#]
RER	0.93 ± 0.03	0.93 ± 0.03	0.92 ± 0.03	0.91 ± 0.04	0.91 ± 0.04	0.00 ± 0.02	−0.01 ± 0.02 [#]	−0.02 ± 0.03	−0.02 ± 0.04
$[La^-]_b$	2.2 ± 0.7	—	—	—	2.5 ± 0.7	—	—	—	0.3 ± 0.8

Note: 0 min refers to the last 90 s of the 5-min reference run prior to the 1 h running session. * $p < 0.05$. ** $p < 0.01$ significant difference within group from rested condition to after 1 h running[#] $p < 0.05$. ^{##} $p < 0.01$ significant difference between groups.

Abbreviations: $[La^-]_b$, lactate concentration in blood; RER, respiratory exchange ratio; VO_2 , oxygen consumption.

should be considered when profiling endurance athletes, as it may influence performance beyond that of VO_{2max} and RE data obtained in a rested condition.

4.1 | Changes in running economy during 1 h running

In line with the hypothesis, VO_2 increased more in AA than in TR. This difference was apparent already after 15 min of running, and continued to increase throughout the 1 h running session. In TR, respiratory exchange ratio (RER) decreased from 0.93 to 0.88. This should, at least in theory, increase VO_2 by 1.2%. Thus, approximately half of the 2.3% increase in VO_2 observed in TR was resultant to a change in substrate utilization from carbohydrate to fat. In AA, RER decreased from 0.93 to 0.91, which should increase VO_2 by 0.5%. In contrast to TR, this can only explain a small fraction of the 6.9% increase in VO_2 , implying that a decrease in mechanical efficiency was more apparent in AA. When energy cost of running was expressed in terms of Kcal min^{-1} , the increase was more than fourfold in AA compared to TR.

The observed changes in RE is within the range of previous reports. In a group of active male students, Sproule⁶ reported 2.6 $mL \cdot kg^{-1} \cdot min^{-1}$ increase in VO_2 after 1 h running at 70% of VO_{2max} . This is similar to the 2.4 $mL \cdot kg^{-1} \cdot min^{-1}$ increase observed in AA following the exact same running protocol. Evidence from cycling experiments involving subjects of similar fitness level is also within the same range (all ~7%), notably with exercise sessions differing slightly in terms of intensity and duration.^{14,22} For TR, direct comparisons to experiments applying the exact same protocol are not available; however, Xu and Montgomery⁵ observed 1.8 $mL \cdot kg^{-1} \cdot min^{-1}$ increase in VO_2 after 90 min of running at 65% of VO_{2max} in a group of trained runners (vs. 1.1 $mL \cdot kg^{-1} \cdot min^{-1}$ after 60 min in TR). Observations along the same line have also been reported among trained cyclists,¹² albeit during longer exercise sessions (~4% increase following 2 h cycling at 65% of VO_{2max}). In general, changes in RE appear to become larger with higher intensity^{5,6} and longer duration of exercise.^{6,13}

4.2 | Changes in attainable VO_{2max} after 1 h running

AA exhibited a decrease in attainable VO_{2max} immediately after the 1 h running session, while TR did not. Whether or not rested values of VO_{2max} are attainable in a fatigued state has been somewhat inconsistent in previous literature, possibly as a result of differences in subject fitness level, exercise modality, testing protocol, and

FIGURE 2 (A) Change in heart rate during 1 h running. (B) Change in RPE during 1 h running. (C) Change in ventilation during 1 h running. (D) Change in breathing frequency during 1 h running. AA, active adults; TR, trained runners. 0 min refers to the last 90 s of the 5-min reference run prior to the 1 h running session. Data are mean $\pm$ SD. $n = 10$ in both groups. * $p < 0.05$, ** $p < 0.01$ significant difference within group from rested condition # $p < 0.05$, ## $p < 0.01$ magnitude of change from rested condition is different between groups.

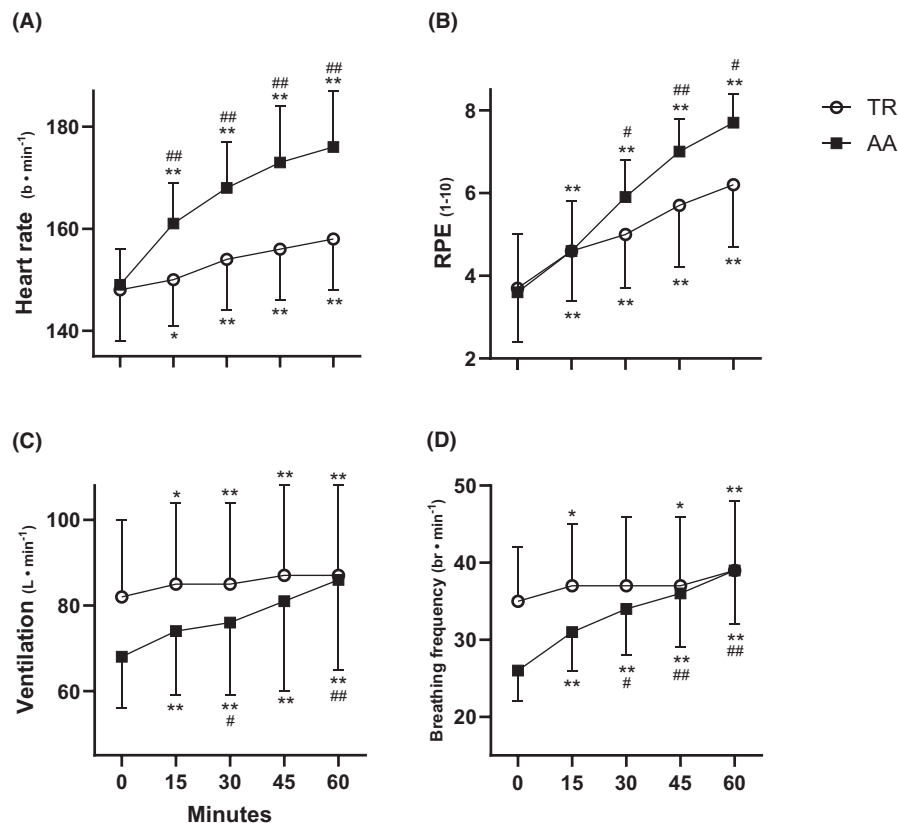


TABLE 4 Ventilatory and performance data from incremental ramp protocols performed in a rested condition, and 2 min after 1 h of running.

	Rested condition	After 1 h	Δ 1 h	$\Delta\%$ 1 h
Trained runners ($n = 10$)				
VO_{2max} ($L \cdot min^{-1}$)	4.47 ± 0.84	4.41 ± 0.84	-0.05 ± 0.11	-1.2 ± 2.4
VO_{2max} ($mL \cdot kg^{-1} \cdot min^{-1}$)	64.4 ± 6.9	63.7 ± 6.8	-0.8 ± 1.6	-1.2 ± 2.5
HR_{max} ($beats \cdot min^{-1}$)	182 ± 7	181 ± 8	-1 ± 2	-0.5 ± 1.2
VE_{max} ($L \cdot min^{-1}$)	157 ± 34	155 ± 33	-2 ± 6	-1.3 ± 3.6
RER_{max}	1.14 ± 0.04	$1.11 \pm 0.02^{**}$	-0.03 ± 0.03	-2.9 ± 2.6
$[La^-]_{b \ max}$	12.2 ± 2.6	10.9 ± 3.2	-1.3 ± 2.6	-10.0 ± 21.2
$Velocity_{max}$ ($km \cdot h^{-1}$)	16.5 ± 1.7	16.3 ± 1.4	-0.2 ± 0.5	-1.3 ± 3.2
Active adults ($n = 10$)				
VO_{2max} ($L \cdot min^{-1}$)	3.93 ± 0.80	$3.72 \pm 0.69^{**}$	$-0.21 \pm 0.15^{\#}$	$-5.0 \pm 3.5^{##}$
VO_{2max} ($mL \cdot kg^{-1} \cdot min^{-1}$)	49.8 ± 5.3	$47.4 \pm 4.2^{**}$	$-2.4 \pm 1.8^{\#}$	$-4.6 \pm 3.5^{\#}$
HR_{max} ($beats \cdot min^{-1}$)	189 ± 6	192 ± 7	$3 \pm 4^{\#}$	$1.4 \pm 2.1^{\#}$
VE_{max} ($L \cdot min^{-1}$)	146 ± 32	143 ± 32	-2 ± 8	-1.7 ± 5.0
RER_{max}	1.17 ± 0.03	$1.11 \pm 0.05^{**}$	-0.05 ± 0.04	-4.5 ± 3.8
$[La^-]_{b \ max}$	12.3 ± 2.3	$10.4 \pm 2.0^{**}$	-1.9 ± 1.7	-14.3 ± 11.7
$Velocity_{max}$ ($km \cdot h^{-1}$)	13.2 ± 1.0	$12.6 \pm 0.6^*$	-0.7 ± 0.7	-4.6 ± 5.9

Note: Data are mean $\pm$ SD. * $p < 0.05$, ** $p < 0.01$ significant difference within group from rested condition to after 1 h running. # $p < 0.05$, ## $p < 0.01$ significant difference between groups.

Abbreviations: $[La^-]_b$, lactate concentration in blood; HR, heart rate; RER, respiratory exchange ratio; VE, ventilation; $Velocity_{max}$, velocity at the end of the VO_{2max} protocol; VO_{2max} , maximal oxygen consumption.

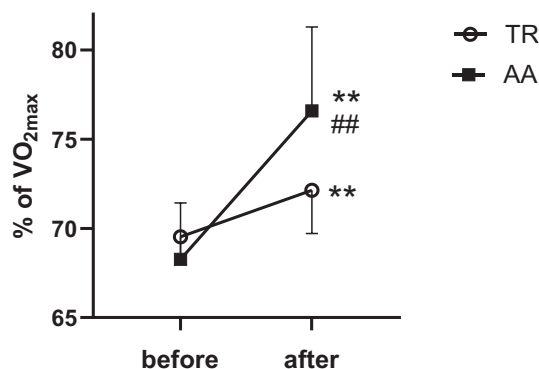


FIGURE 3 Change in relative intensity (submaximal VO_2 /attainable $\text{VO}_{2\text{max}}$) from rested condition to the end of 1 h running. Data are mean $\pm$ SD. $n = 10$ in both groups. ** $p < 0.01$ change within group from rested condition to the end of 1 h running. ## $p < 0.01$ magnitude of change is different between groups.

degree of exhaustion. In endurance athletes of approximately the same level as TR, Dressendorfer⁷ and De Vito et al.⁸ observed 4%–7% decrements in attainable running $\text{VO}_{2\text{max}}$ 30 min after completion of a half marathon⁷ and immediately after the first two segments of an Olympic triathlon.⁸ While TR did not exhibit such a decrement, the fatiguing exercise bout in Dressendorfer and DeVito et al. were of longer duration and higher intensity than in current study, which likely explains the discrepancy. In contrast to the studies by Dressendorfer and DeVito et al., Clark et al.^{12–14} observed no differences in cycling $\text{VO}_{2\text{max}}$ when tested during 3 min of all-out effort, before and after 2 h of cycling at approximately 65% of $\text{VO}_{2\text{max}}$. In these studies, when $\text{VO}_{2\text{max}}$ was assessed by longer continuous effort above critical power, it was lower in the fatigued state, indicating that a short and vigorous testing protocol may be required to reach maximal values in the fatigued state. Given the distinct difference between TR and AA in the current study, comparisons to the Clark studies are not straightforward, as the subjects in two of these three studies were trained cyclists¹² and athletes.¹³ Interestingly, in the third study by Clark et al.,¹⁴ nonathlete male subjects, comparable to AA, attained similar cycling $\text{VO}_{2\text{max}}$ after 2 h of cycling. This raises the question whether prolonged running may lead to larger drift in physiological characteristics than prolonged cycling.⁴ Arguably, the repetitive foot strikes during running may have larger impact on the working musculature, possibly leading to muscle damage,²³ especially in subjects that are not used to prolonged running, such as AA. Exercising in the heat (35°C), Wingo, Stone and Ng⁹ observed similar decrements in $\text{VO}_{2\text{max}}$ after 45 min of running and cycling at 60% of $\text{VO}_{2\text{max}}$, advocating that this may not be the case. It is nevertheless possible that testing modality may influence “ $\text{VO}_{2\text{max}}$ resilience,” as whole-body $\text{VO}_{2\text{max}}$ is lower (~6%–9%) when tested during cycling than during

running,^{24,25} unless subjects are trained cyclists. Running $\text{VO}_{2\text{max}}$, which more accurately represents the upper limit of whole body $\text{VO}_{2\text{max}}$, may therefore be compromised at a lower degree of fatigue than cycling $\text{VO}_{2\text{max}}$, in particular if central factors that compromise cardiac output are subject to fatigue.

4.3 | Changes in relative intensity and performance during 1 h running

The combined changes in RE and $\text{VO}_{2\text{max}}$ resulted in a larger increase in relative intensity in AA than in TR. This difference was mirrored by (and correlated with) a larger increase in HR in AA than in TR, during the 1 h running session. Wingo, Lafrenz, Ganio, Edwards, and Cureton²⁶ reported that the magnitude of cardiovascular drift during prolonged cycling was related to the magnitude of changes in $\text{VO}_{2\text{max}}$ and relative intensity, albeit in a heated environment. From a practical point of view, if HR monitoring during steady pace exercise has potential to predict changes in $\text{VO}_{2\text{max}}$ and RE, it is possible that some primitive assessment of resilience could be made available outside advanced testing laboratories. To the authors best knowledge, this study is the first to show that the magnitude of cardiovascular drift may be related to training status. It is likely that the difference in ΔHR is related to superior ability to resist changes in $\text{VO}_{2\text{max}}$ and RE in TR compared to AA.

Interestingly, the change in relative intensity was not only different between TR and AA; also within the groups large differences were observed. Among AA the increase in relative intensity ranged from 1% to 14%, while TR ranged from 0% to 5%. Strikingly, one of the subjects in AA exhibited similar resilience as the best TR subjects, despite having the lowest $\text{VO}_{2\text{max}}$ in the study (~44 mL kg⁻¹ min⁻¹). This issue is highlighted by Jones,⁴ reporting that the fatigue-induced change in critical power in the three studies by Clark et al.^{12–14} ranged from 0.4% to 32% between the subjects, with no correlation to $\text{VO}_{2\text{max}}$. The great interindividual differences in resilience, even within quite homogenous groups, illustrate that the determinants of physiological resilience are not fully understood, and also confirms resilience as an independent physiological variable, as suggested by Jones.⁴

In the current study, velocity at the end of the $\text{VO}_{2\text{max}}$ stepwise protocol was the only measure of running performance. In agreement with the physiological resilience data, AA terminated the $\text{VO}_{2\text{max}}$ protocol at a lower velocity after 1 h running, while TR exhibited no change in performance. Although recognizing that this measure of performance is not the most sensitive, the strong correlation between *relative intensity at the end of 1 h running*, and *change in running performance*, demonstrates

the performance-wise significance of fatigue-induced changes in $\text{VO}_{2\text{max}}$ and RE. In agreement with our observations, Clark et al.,^{12–14} reported that % of $\text{VO}_{2\text{max}}$ at the end of the fatiguing exercise session was the single variable that correlated strongest with fatigue-induced change in performance during 3 min all-out cycling. Perhaps unsurprising, avoiding to drift into very high relative intensity during prolonged steady pace exercise, appears to conserve performance during a few minutes of all out exercise, such as for example the last kilometer of a marathon.

4.4 | Determinants of physiological resilience

Although TR exhibited superior physiological resilience compared to AA, the determinants of this difference are difficult to deduct from the current data. During the 1 h running session, $[\text{La}^-]_b$ was somewhat higher in AA than in TR. While this difference could suggest that AA were exercising in a higher intensity domain than TR, the difference in $[\text{La}^-]_b$ was present also during the easy warm up procedures. An increase in $[\text{La}^-]_b$ above warm up levels was apparent at “min 0” in both TR and AA ($p < 0.01$ in both groups), and the magnitude of increase was not different between the groups (TR: $0.7 \pm 0.5 \text{ mmol L}^{-1}$; AA: $0.9 \pm 0.7 \text{ mmol L}^{-1}$, $p = 0.38$). Indeed, lower $[\text{La}^-]_b$ are observed at metabolic thresholds with increasing aerobic capacity.²⁷ In agreement with this interpretation, gas exchange threshold occurred at similar percentage of $\text{VO}_{2\text{max}}$ in the two groups, and below 70% of $\text{VO}_{2\text{max}}$ in all subjects, advocating that all subjects exercised in the heavy-intensity domain from the beginning of the 1 h running session. Exercising in the heavy-intensity domain is associated with a VO_2 slow component, leading to a slight increase in VO_2 over time.²⁸ Indeed, such an increase was observed in both TR and AA, but the magnitude of increase was larger in AA. With regard to the upper end of the heavy-intensity domain, the lack of change in $[\text{La}^-]_b$ during 1 h running indicates that both TR and AA were below maximal lactate steady state^{29,30} and critical speed.³¹ However, it is still possible that AA were closer to the upper boundary of the heavy-intensity domain than TR, as could be indicated by the lower percentage of $\text{VO}_{2\text{max}}$ at 2 and 3 mmol L^{-1} $[\text{La}^-]_b$ observed in AA. Even when percentage of $\text{VO}_{2\text{max}}$ was compared at warm up $[\text{La}^-]_b + 1.5 \text{ mmol L}^{-1}$, which is probably a more fair comparison between the groups, a non-significant ~2% difference was apparent in favor of TR. For sure, VE, BF, HR, and RPE development during the 1 h running session indicate that TR were closer to physiological steady state than AA.

Based on weight loss and fluid intake, dehydration was similar between groups. Carbohydrate availability has been

introduced as a possible determinant of physiological resilience.^{4,13} In the current study no significant differences were observed between groups in RER development throughout the 1 h running session, or at the end of the $\text{VO}_{2\text{max}}$ protocols. Both groups exhibited a decrease in $[\text{La}^-]_{b\text{max}}$ after the incremental stepwise protocol in the fatigued condition, which is likely to reflect glycogen depletion. However, this decrease was not different between the groups.

Changes in VE, BF, and HR correlated with changes in VO_2 , $\text{VO}_{2\text{max}}$ and relative intensity, and were considerably larger in AA than in TR (ΔHR 18% vs. 7%, ΔVE 28% vs. 5%, ΔBF 50% vs. 11%). While the O_2 -cost of increased HR is limited, the increased ventilation has larger potential to impact VO_2 , and may possibly explain part of the changes in VO_2 during 1 h running, especially in AA. Although the increase in ventilation is likely secondary to an increased metabolic demand of running, the combination of decreased mechanical efficiency and a corresponding increase in ventilation may amplify the increase in VO_2 during prolonged exercise in subjects with poor resilience.

Given the lack of relation to any obvious physiological determinant, the most likely explanation of the difference in resilience between TR and AA remains the exposure to systematic running training. In the current study a weekly training load of *69 km running + 3.4 h other endurance training* was associated with superior resilience than *9 km running + 0.8 h other endurance training*. Field data from elite cyclists indicate that resilience may continue to increase up to the highest levels, with even more extensive and long-term exposure to endurance training.¹⁷ On the lower end of the scale, the magnitude and type of training required to induce improvements in physiological resilience have yet to be explored, beyond the fact that the training exposure of TR in the current study appears to be sufficient to improve resilience.

4.5 | Study limitations

The somewhat “old school” approach used to normalize exercise intensity between subjects in the current study, using a fixed % of $\text{VO}_{2\text{max}}$, has been problematized in recent literature.^{32,33} Specifically, this method does not take into account that the boundaries between different intensity domains (i.e., moderate, heavy, and severe) occur at different % of $\text{VO}_{2\text{max}}$ between subjects. As a consequence of this, physiological responses to a given % of $\text{VO}_{2\text{max}}$ may be quite heterogeneous, especially if subjects are exercising in different intensity domains. Although gas exchange threshold data indicates that 70% of $\text{VO}_{2\text{max}}$ was in the heavy domain for all subjects, it cannot be excluded that AA were closer to their maximal lactate steady state

than TR. In turn, such higher metabolic strain in AA may explain some of the observed differences in resilience between AA and TR.

5 | CONCLUSION

During 1 h running at 70% of $\text{VO}_{2\text{max}}$, VO_2 increased more in AA than in TR, while attainable $\text{VO}_{2\text{max}}$ decreased in AA and remained unchanged in TR. As a consequence of this, relative intensity increased more in AA than in TR. These findings imply that TR exhibit superior ability to resist changes in RE and $\text{VO}_{2\text{max}}$ compared to AA, which may indicate that long term systematic running training increases physiological resilience.

5.1 | Perspective

For decades, the three-headed model consisting of maximal oxygen consumption, lactate threshold, and work economy¹⁵ has constituted the foundation of endurance physiology research. Arguably, as suggested by Jones,⁴ physiological resilience may compliment this model as a fourth parameter, as the three original components are susceptible to change during prolonged exercise. The fact that TR are superior in resisting changes in $\text{VO}_{2\text{max}}$ and RE compared to AA, may indicate that physiological resilience is a trait that adapts to training, although longitudinal experiments are required to confirm this. Being a relatively new area of research, knowledge of the factors that influences physiological resilience are sparse. Knowledge of which training exposure, or for that matter factors other than training that influences physiological resilience, would be useful for anyone striving to improve endurance performance.

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CONFLICT OF INTEREST STATEMENT

The author reports no conflict of interest.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

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